STUDENT NAME:	Francisco Sandoval Hernández
SUPERVISOR NAME:	Moisés Martínez Muñoz

TUTORING SESSION Nº:

☐ 1. Start
☒ 2. State of the Art / Previous Research
☐ 3. Project Plan
☐ 4. Project Development
☐ 5. Technical Work Closure
☐ 6. Report Review
☐ 7. Other: Specify

DATE: 22/01/2026
START: 19:00
END: 20:00

DISCUSSED TOPICS:

During this tutoring session, the initial research and technical background of the project were reviewed. The discussion focused on the main families of techniques that could be useful for sign language recognition: convolutional neural networks, object detection models, recurrent neural networks, LSTM and GRU architectures, attention mechanisms, Transformer-based models and landmark-based pose estimation.

The first technical experiments were also discussed. The student presented the initial work on hand detection using datasets such as EgoHands and the Kaggle Hand Dataset, together with early static ASL letter-recognition experiments using Sign Language MNIST and ASL Alphabet. These experiments showed that hand localization and static letter classification were feasible but also revealed that they did not fully address the main challenge of recognizing complete dynamic signs.

A methodological pivot was considered. Instead of continuing only with a pixel-based route centred on static letters, the supervisor and student discussed the possibility of using MediaPipe hand landmarks as a compact representation of hand motion. This approach was considered more appropriate for word-level sign recognition because it reduces background and appearance noise and provides a structured temporal input for sequence models.

The available word-level datasets were reviewed, especially WLASL, as a possible benchmark for dynamic sign recognition. It was agreed that the project should test a small and controlled subset first, to reduce training time and allow comparison between different sequence models under the same conditions.

It was agreed that the next work should focus on comparing the initial pixel-based route with a landmark-based approach. The student would continue testing hand-detection and static-recognition experiments, while also preparing the first WLASL subset for word-level recognition.

ACUERDOS PARA LA SIGUIENTE REUNIÓN:

Item	Agreement / Task	Person in Charge	Deadline
1	Continue the exploratory experiments with hand detection and static ASL letter recognition.	F.S.H	06/02/2026
2	Analyze whether MediaPipe landmarks could be used as a better representation for dynamic signs.	F.S.H	06/02/2026

3	Prepare an initial WLASL subset for word-level ASL recognition.	F.S.H	06/02/2026
4	Compare possible temporal architectures such as CNN-LSTM, LSTM, GRU and Transformer-based models.	F.S.H	06/02/2026

SCHEDULED DATE OF THE NEXT MEETING: **06/02/2026 – 19:00**

Supervisor's Comments:

Supervisor's Signature:	Student's Signature:
 Moisés Martínez Muñoz	